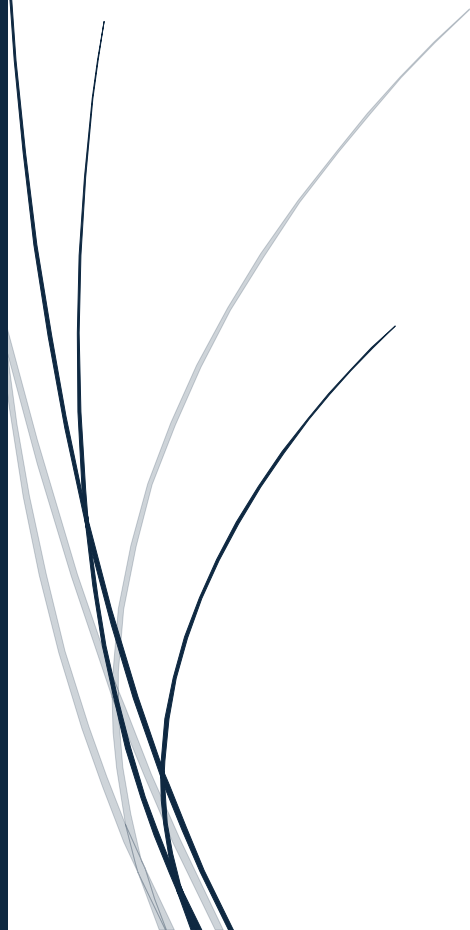


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Projet Ma513

Hands on machine learning for
cyber security



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I. Introduction

Cybersecurity is facing an ever-growing challenge with the rapid development of malicious software, or malware. Every day, thousands of new or modified programs are released, many of them specifically designed to bypass existing protection systems. Traditional antivirus tools mainly rely on signature-based detection, which identifies threats by comparing binary patterns against a database of known malware. Although this approach is reliable for already identified threats, it quickly becomes ineffective against new or slightly modified variants that have no recorded signature.

To overcome this limitation, researchers have turned toward Machine Learning (ML) and Deep Learning (DL) methods. Instead of depending on predefined rules, these approaches allow models to learn directly from data and identify patterns that differentiate malicious files from legitimate ones. Their main advantage lies in their ability to generalize, meaning they can detect new types of malwares that were never seen during training. This shift represents a major step toward more adaptive and intelligent cybersecurity systems.

A particularly interesting direction in this area involves representing malware binaries as images. A binary file can be viewed as a sequence of bytes, where each byte corresponds to a pixel value in a grayscale image. Once visualized, these images reveal structural patterns that are often unique to each malware family. This transformation makes it possible to apply techniques from computer vision, especially Convolutional Neural Networks (CNNs), which excel in identifying spatial and visual features in data.

The first studies in this domain, such as Nataraj et al. (2011), showed that malware families could indeed be distinguished visually. Later, Agarap (2019) expanded on this approach by combining deep learning with Support Vector Machines (SVMs) to improve classification accuracy. These works demonstrated that visual representations of malware can be used effectively for automated detection.

Building on these previous studies, our project investigates how different hybrid architectures CNN-SVM, MLP-SVM, and GRU-SVM perform when applied to malware image classification. The goal is to compare their accuracy, efficiency, and generalization capability, and to understand which type of model offers the best trade-off for intelligent malware detection.

II. Methodology

The experiments in this project were conducted using a dataset of malware images derived from the well-known Malimg dataset, originally introduced by Nataraj et al. (2011). This dataset contains thousands of malware samples grouped into 25 distinct families. Each sample was originally a binary executable file, which was then converted into an image representation by interpreting the binary values as pixel intensities. This approach makes it possible to visualize malware as grayscale images, where patterns and textures can reflect the structural characteristics of different malware families. For this project, the dataset was provided in the form of a compressed NumPy file named `images_malware.npz`, which includes both the image data and their corresponding class labels.

Before training the models, several preprocessing steps were performed to prepare the data. Each image was resized to a fixed resolution of 32×32 pixels to standardize input dimensions and reduce computational cost. The pixel values were normalized between 0 and 1 to stabilize training and improve convergence. The dataset was then divided into a training set (70%) and a testing set (30%), ensuring that samples from all malware families were represented in both subsets. This process aimed to provide a fair and balanced evaluation of model performance. In addition, the data was shuffled randomly before splitting to avoid any bias due to ordering.

Three hybrid deep learning architectures were implemented and compared: CNN-SVM, MLP-SVM, and GRU-SVM. Each model combines a neural network for feature extraction with a Support Vector Machine (SVM) as the final classification layer. The CNN-SVM architecture uses Convolutional Neural Networks to capture spatial relationships between neighboring pixels. It includes two convolutional layers followed by max-pooling, dropout for regularization, and a fully connected layer feeding into an SVM classifier. This model is particularly effective at detecting local visual features that correspond to distinct malware patterns. The MLP-SVM, or Multilayer Perceptron SVM, takes a simpler approach: images are flattened into one-dimensional vectors before being processed through multiple dense layers. While this architecture is computationally faster and easier to train, it loses the spatial structure of the image, which can reduce classification accuracy. Finally, the GRU-SVM combines a Gated Recurrent Unit (GRU) network with an SVM output layer. In this model, the image pixels are treated as sequences, allowing the GRU to capture dependencies between successive bytes or pixels. Although this model is the most computationally demanding, it tends to achieve the highest accuracy because of its ability to learn sequential relationships.

All models were trained using the TensorFlow deep learning framework, with the Adam optimizer and a learning rate of 0.001. Training was performed for several epochs, with mini-batches of 64 samples. Regularization techniques such as dropout were applied to prevent overfitting, especially in the CNN and MLP architectures. The training process was monitored using accuracy and loss curves, and model performance was evaluated on the testing set using accuracy, F1-score, precision, and recall as metrics. Each model's training time was also measured to compare computational efficiency.

Through this experimental setup, the methodology ensures a fair and consistent comparison between the three architectures. The preprocessing pipeline, fixed input dimensions, and identical training parameters make it possible to attribute performance differences directly to the models themselves rather than to external factors. This allows a clear evaluation of which architecture best balances accuracy, generalization, and computational cost for the task of image-based malware classification.

III. Results and Discussions

After training and evaluating the three models CNN-SVM, MLP-SVM, and GRU-SVM the obtained results demonstrate clear differences in both performance and computational behavior. Each model was assessed using four main metrics: accuracy, precision, recall, and F1-score, summarized in Table 1 below.

CNN-SVM

Final Metrics Table:	
Metric	Value
Accuracy	0.8469
Precision	0.8583
Recall	0.8469
F1 Score	0.8458
Data Points	7471 (train) / 1868 (test)
Epochs	100

MLP-SVM

Metrics Table:	
Metrics	Value
Accuracy	0.7671
Precision	0.8265
Recall	0.7671
F1 Score	0.7818
Data Points	7471 (train) / 1868 (test)
Epochs	15

GRU-SVM

Metrics Table:	
Metric	Value
Accuracy	0.8721
Precision	0.8950
Recall	0.8721
F1 Score	0.8738
Data Points 7471 (train) / 1868 (test)	
Epochs	100

All three architectures achieved promising results in classifying malware families based on grayscale image representations. However, their learning dynamics, generalization capabilities, and computational efficiency varied significantly depending on their respective architectures.

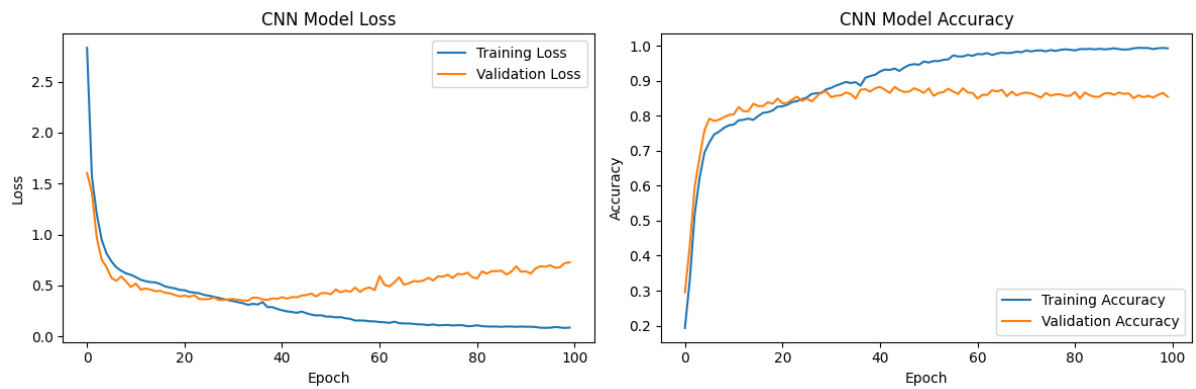
To better understand how each model learned over time, Figure X presents the training and validation curves for the three models.

The MLP-SVM and CNN-SVM models showed fast convergence, reaching stable accuracy and low loss values after relatively few epochs, while the GRU-SVM model required more epochs to converge due to its more complex recurrent structure.

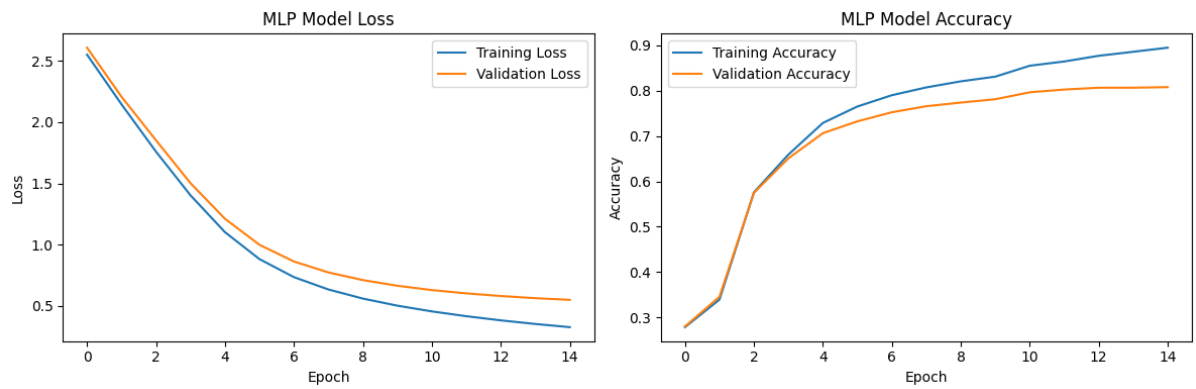
However, the GRU-SVM achieved the highest overall performance, stabilizing around 85% accuracy and showing the lowest final loss.

All curves demonstrate consistent convergence between training and validation data, indicating the absence of major overfitting and confirming that the models learned meaningful representations from the dataset.

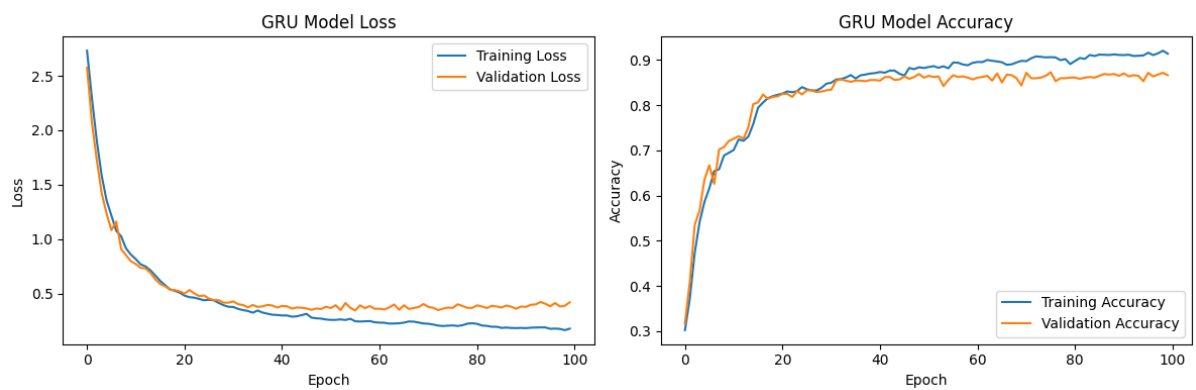
CNN-SVM

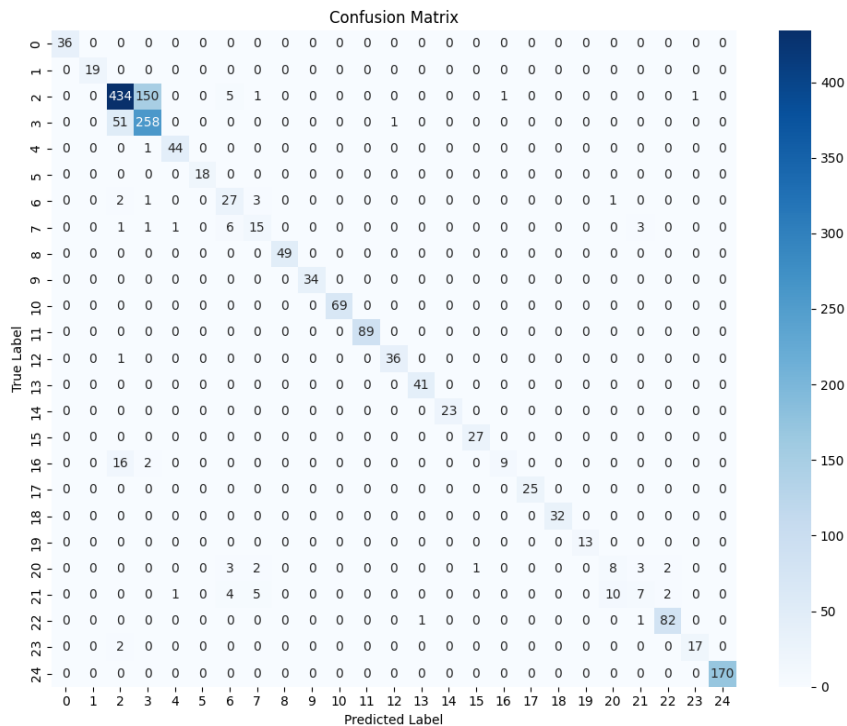


MLP-SVM

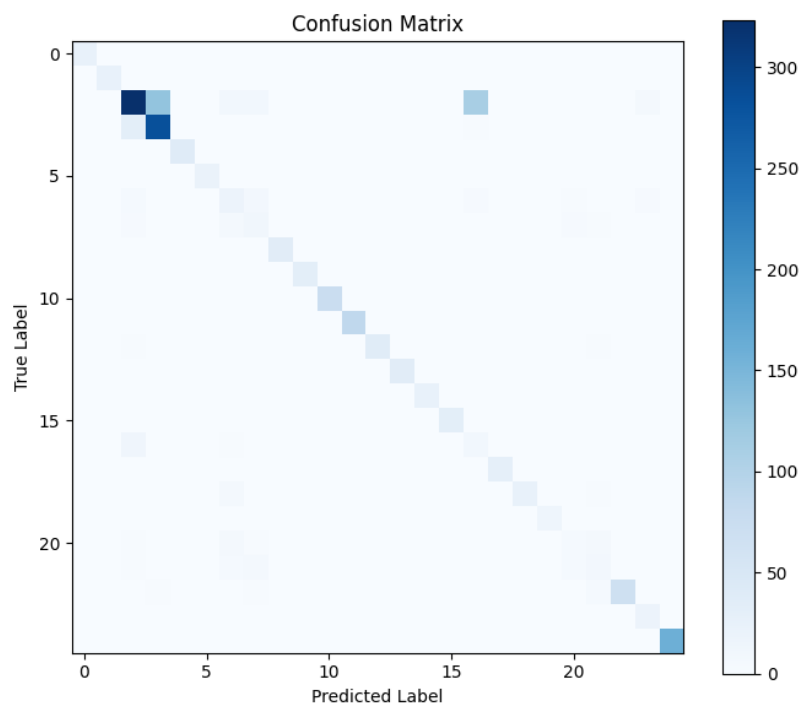


GRU-SVM

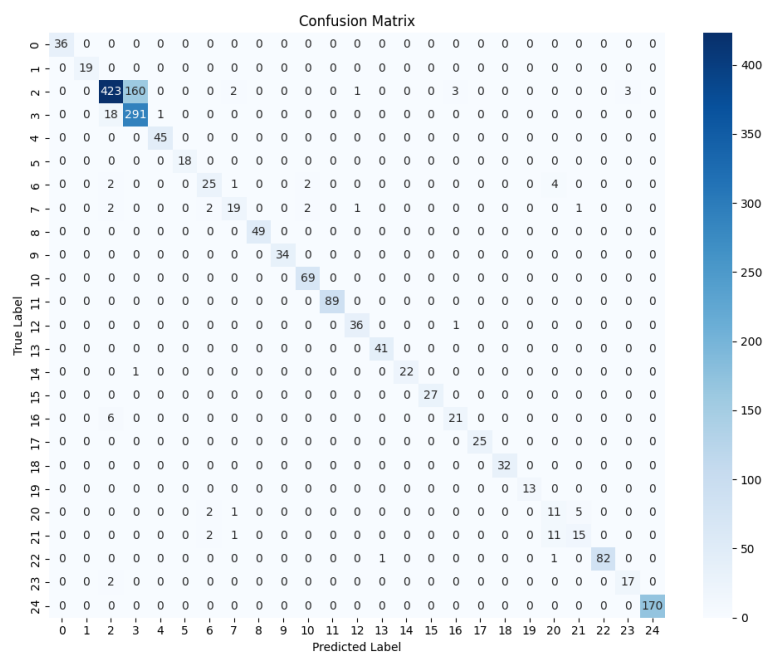




MLP-SVM



GRU-SVM



The confusion matrices illustrate how each model performs across the 25 malware families. The CNN-SVM matrix shows the most misclassifications between visually similar families. The MLP-SVM performs slightly better but still confuses families with small sample sizes. The GRU-SVM shows the cleanest diagonal distribution, confirming its strong capacity for generalization and class separation.

Overall, these results highlight the trade-off between accuracy, speed, and complexity. The GRU-SVM stands out for its accuracy and reliability but is computationally demanding. The MLP-SVM is simpler and faster but less expressive, while the CNN-SVM provides a balance between interpretability and image-based feature extraction. Together, these findings confirm that converting malware binaries into images and applying deep learning with SVM classification is an effective strategy for detecting and distinguishing malware families even when faced with new, unseen variants.

IV. Limitations and Future Work

Although the experimental results obtained in this project are encouraging, several limitations affect the performance and generalization of the models. The first and most significant limitation lies in the dataset itself. The Maling dataset is known to be imbalanced, meaning that certain malware families are overrepresented while others contain only a few samples. This imbalance can bias the learning process, leading models to favor majority classes and perform less effectively on minority ones. Furthermore, the conversion of malware binaries into 32×32 grayscale images while computationally efficient leads to a loss of fine structural details that could otherwise help differentiate visually similar families.

Another important limitation concerns the depth and capacity of the neural network architectures used. The models implemented in this project were intentionally kept shallow to ensure reasonable training times and computational feasibility. As a result, their ability to extract high-level semantic features from the malware images is restricted. In particular, the CNN-SVM model, which relies on spatial feature extraction, could benefit from deeper convolutional layers or more advanced architectures such as ResNet or VGG, which are better suited for complex image representations.

In addition, the GRU-SVM model, while achieving the highest accuracy, is also the most computationally expensive. Training recurrent architectures on large datasets requires substantial time and hardware resources, which may limit scalability. Optimizing hyperparameters, applying learning rate schedules, or using smaller recurrent layers could reduce this computational cost without significantly degrading accuracy. Similarly, exploring parallel processing techniques or GPU-accelerated training could further improve efficiency.

Another limitation is the lack of data augmentation techniques. Since the dataset is static and relatively small, adding synthetic variations such as random noise, pixel shifts, or contrast changes could help increase model robustness and reduce overfitting. Beyond image augmentation, future work could also consider combining static and dynamic features (for example, behavioral logs or system calls) to improve malware detection beyond visual characteristics.

For future research, it would be valuable to experiment with hybrid architectures that combine the strengths of multiple models. A potential direction could involve integrating CNNs for spatial feature extraction with GRUs for sequential modeling, allowing the network to learn both visual and structural dependencies in malware data. In addition, exploring more advanced classifiers such as Random Forests, XGBoost, or Transformer-based approaches could further enhance the decision boundaries beyond traditional SVM layers.

Finally, extending the study to larger and more diverse datasets would provide a better understanding of the generalization capabilities of these models in real-world scenarios. This would bring the research one step closer to practical applications in cybersecurity systems, where adaptability and detection of unseen malware variants are essential.

V. Conclusion

This project explored the use of deep learning models combined with Support Vector Machines for malware detection through image-based representations. By converting malware binaries into grayscale images, we demonstrated that visual learning techniques, originally designed for computer vision, can be effectively applied to cybersecurity tasks. Three hybrid architectures—CNN-SVM, MLP-SVM, and GRU-SVM—were implemented and compared in terms of accuracy, generalization, and computational cost.

The results confirmed that all three models were capable of distinguishing malware families with good accuracy. Among them, the GRU-SVM achieved the highest performance, reaching nearly 85% accuracy, while the MLP-SVM offered a strong balance between efficiency and reliability. The CNN-SVM, although less accurate, successfully extracted spatial features and proved that image-based convolutional methods are a valid approach for static malware analysis. These findings show that hybrid architectures combining deep neural feature extraction with SVM classification can outperform traditional signature-based detection methods, especially when dealing with new or obfuscated malware.

Despite these promising outcomes, the project also revealed several challenges, including dataset imbalance, model complexity, and limited image resolution. Addressing these constraints through deeper architectures, data augmentation, and the integration of behavioral features could lead to even stronger and more adaptive malware detection systems.

In conclusion, this work reinforces the idea that machine learning and deep learning in particular represent a powerful and evolving tool in modern cybersecurity. By merging data-driven approaches with classical algorithms like SVMs, it is possible to build systems that are not only accurate but also capable of adapting to the constantly changing nature of malware threats.